

Indoor Air Quality Monitoring

Impact of FDM 3D Printing and Filtration in a Residential Setting

A two-phase self-hosted sensor study · Bambu Lab P2S · Sensirion SPS30 / SGP41

Executive Summary

This report analyses indoor air quality data collected with a custom-built monitoring station, alongside synchronised operating data from a Bambu Lab P2S 3D printer, in a flat where the printer sits near the kitchen and a window. It covers two phases: an initial 25-day particulate study (16 May – 11 June 2026), and a later gas-phase study (25 July – 17 August 2026) made possible after firmware and hardware faults affecting the gas sensor were diagnosed and fixed.

The key findings are:

- Printing PLA or PETG had little measurable effect on PM2.5, with or without filtration. Within each period, particulate levels while printing were statistically indistinguishable from the room baseline when the printer was off.
- The apparent benefit of the VentoBox on PM2.5 could not be isolated. The room baseline itself was lower in the (later) VentoBox period than in the (earlier) no-filter period — even with the printer off — pointing to changing outdoor conditions rather than the filter. The filter's real value likely lies in gases and ultrafine particles, which this sensor does not measure (see Sections 4 and 7).
- The largest PM2.5 spikes were not caused by the printer. They cluster in the morning and evening and are consistent with cooking and/or outdoor air entering through the window — expected given the sensor's location.

The decisive finding came only after the gas sensor was repaired: VOC rises during printing (median index 196 while printing versus 118 idle, $p \approx 7 \times 10^{-14}$) — a signal the particulate sensor missed entirely. The VOC Index is a relative measure, not a concentration, so this shows that printing changes the air measurably; it does not establish a health risk, and no WHO guideline exists for VOCs as a category. NOx stayed at baseline throughout, as expected for FDM printing. See Section 7.

1. Background & Objective

After upgrading from an old RF100 (PLA only) to a Bambu Lab P2S and beginning to print PETG, the motivation was simple: understand what is actually being breathed in a small flat during printing. Commercial air-quality monitors with charting and alerting were expensive, and inexpensive units have a reputation for unreliable readings — so a custom station was built around research-grade Sensirion sensors.

The system streams sensor data over MQTT to a self-hosted stack (Mosquitto, Telegraf, InfluxDB, Grafana) running on a Raspberry Pi. Crucially, the same pipeline also ingests live status from the Bambu Lab P2S over its local MQTT API, allowing air-quality readings to be correlated directly with printer activity, temperatures, and filtration settings.

2. Methodology

2.1 Hardware

- ESP32 (WROOM-32U, external antenna) — microcontroller and WiFi link
- Sensirion SPS30 — laser particulate matter sensor (PM1, PM2.5, PM4, PM10)
- Sensirion SGP41 — metal-oxide gas sensor (VOC / NOx index)
- Bambu Lab P2S — print status via local MQTT (nozzle/bed temperature, print state, VentoBox fan speed)

2.2 Data collection

Readings were published every few seconds and aggregated to 5-minute means for analysis. The dataset spans 16 May to 11 June 2026 (25 days, 6,781 five-minute samples). Sensor and printer streams were time-aligned to the nearest sample. "Active printing" was defined as nozzle temperature above 150 °C, a reliable indicator of extrusion.

A small number of sensor-startup artefacts (simultaneous zero readings across all channels) were removed prior to analysis.

3. Overall Air Quality

Across the full monitoring period, indoor PM2.5 was generally good by WHO standards:

Metric	Value
Mean PM2.5	5.7 µg/m ³
Median PM2.5	4.5 µg/m ³
95th percentile	12.8 µg/m ³
Maximum (5-min mean)	372 µg/m ³
Time above WHO 24h guideline (15 µg/m ³)	3.0%
Time above alarm threshold (35 µg/m ³)	0.31%

For context, the WHO 2021 air quality guideline for PM2.5 is an annual mean of 5 µg/m³ and a 24-hour mean of 15 µg/m³. The flat sat below the 24-hour guideline 97% of the time. The rare excursions are examined in Section 5.

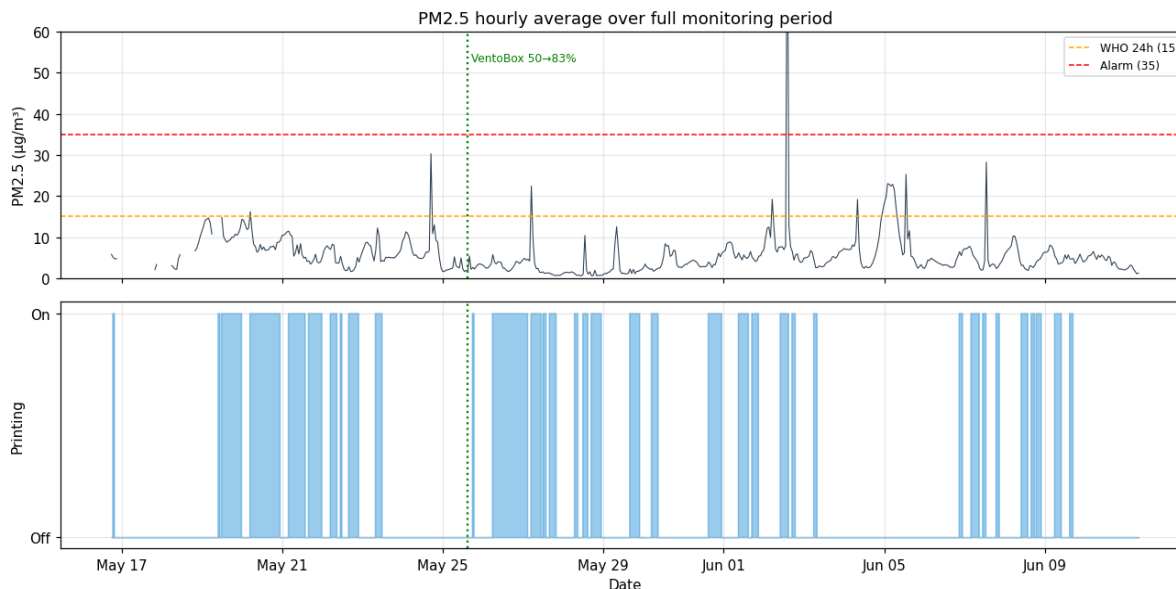


Figure 1. Hourly-averaged PM2.5 over the full period (top) with printing activity (bottom). The green dotted line marks the VentoBox installation on 25 May. Orange and red dashed lines are the WHO 24-hour guideline and the chosen alarm threshold.

4. The VentoBox: What the Data Can and Cannot Show

The VentoBox filtration unit was installed on 25 May. Before that date the printer ran with no filtration. A natural question is whether the filter reduced printing-related particulates. At first glance the data appears to say yes: median PM2.5 during printing was $6.7 \mu\text{g}/\text{m}^3$ before installation and $2.9 \mu\text{g}/\text{m}^3$ after. However, this comparison is confounded and does not support a causal claim.

The problem is that the room's background air quality also changed between the two periods — for reasons unrelated to the printer or the filter. Looking only at times when the printer was switched off, the median room baseline was $6.1 \mu\text{g}/\text{m}^3$ during the no-filter period (mid-May) and $3.6 \mu\text{g}/\text{m}^3$ during the VentoBox period (late May to June). Since the printer (and therefore the VentoBox) was off during these measurements, the filter cannot explain the drop. The most likely cause is changing outdoor conditions — mid-May included several visibly polluted days (daily baselines of $12\text{--}13 \mu\text{g}/\text{m}^3$ on 19–21 May) while early June was cleaner.

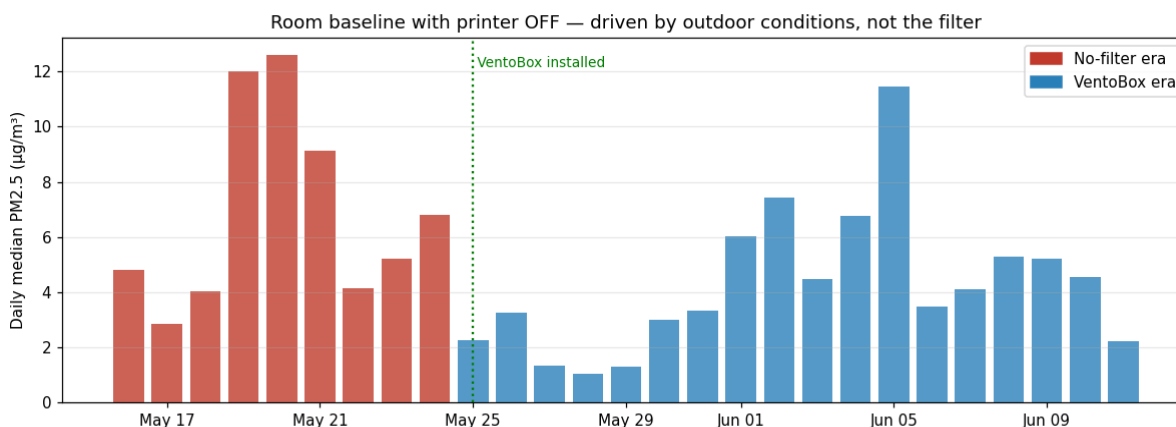


Figure 2. Daily median PM2.5 with the printer switched off. Because the printer and filter were inactive, this curve reflects the room's background air — which was higher in mid-May and lower in June, independent of the VentoBox.

The honest way to assess the printer's impact is to compare printing against the room baseline within the same period, so that outdoor conditions are held roughly constant. Doing this shows printing had a minimal effect in both periods:

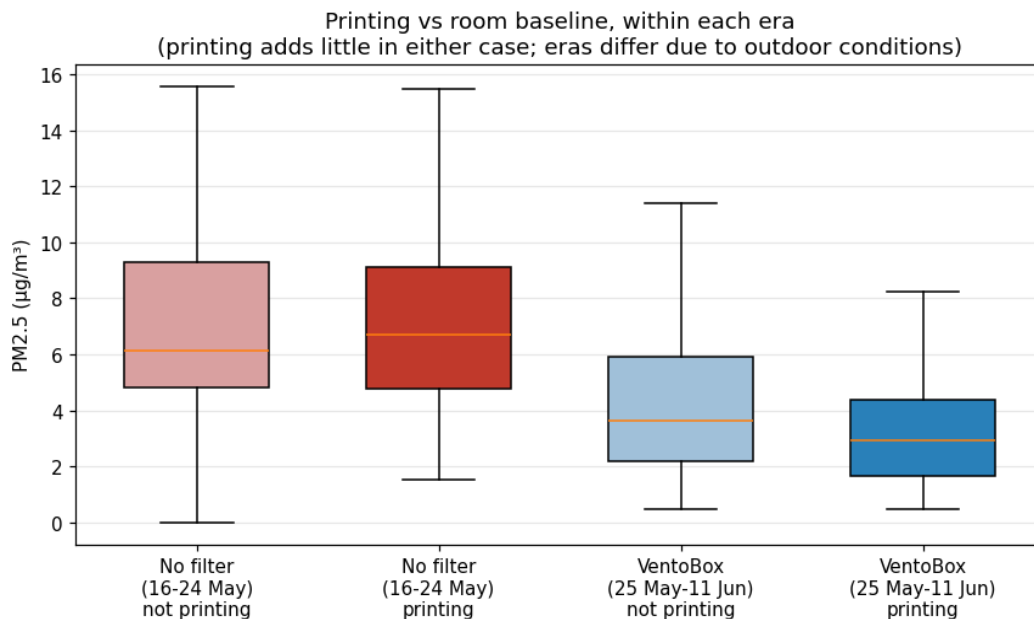


Figure 3. PM2.5 while printing versus the room baseline, within each period. In both cases the printing and not-printing distributions overlap heavily; the visible shift is between periods (outdoor conditions), not between printing and idle.

In the no-filter period, printing raised median PM2.5 by just +0.6 µg/m³ over baseline — not statistically significant (Mann-Whitney $p = 0.18$). In the VentoBox period, printing was actually slightly below baseline. In neither case did printing meaningfully degrade the air.

Conclusion on the VentoBox: this dataset cannot demonstrate a PM2.5 benefit from the filter, because printing barely raised particulates even without it. This does not mean the VentoBox is useless. Two important caveats apply. First, the SPS30 measures particles down to roughly 300 nm and does not capture the ultrafine particles (UFPs, <100 nm) that dominate FDM emissions — a HEPA stage can still help here. Second, the filter's activated-carbon layer targets gases (VOCs, styrene), which were not measured in this study. The filter may well provide benefits this particulate sensor is simply blind to.

5. PLA vs PETG: Does Material Matter?

Although filament type was not logged for each print, nozzle and bed temperatures act as a reliable proxy. The printing data fell into two clean clusters: a low-temperature group (nozzle ~220 °C, bed 55 °C) consistent with PLA, and a high-temperature group (nozzle 245–255 °C, bed 70 °C) consistent with PETG. Using these signatures, 93% of printing samples could be classified confidently.

A fair comparison requires controlling for confounders: PLA was printed more often before the VentoBox was installed, and the dataset contains occasional external pollution spikes unrelated to printing. The comparison below is therefore restricted to the period after VentoBox installation, with external events (PM2.5 > 50 µg/m³) excluded.

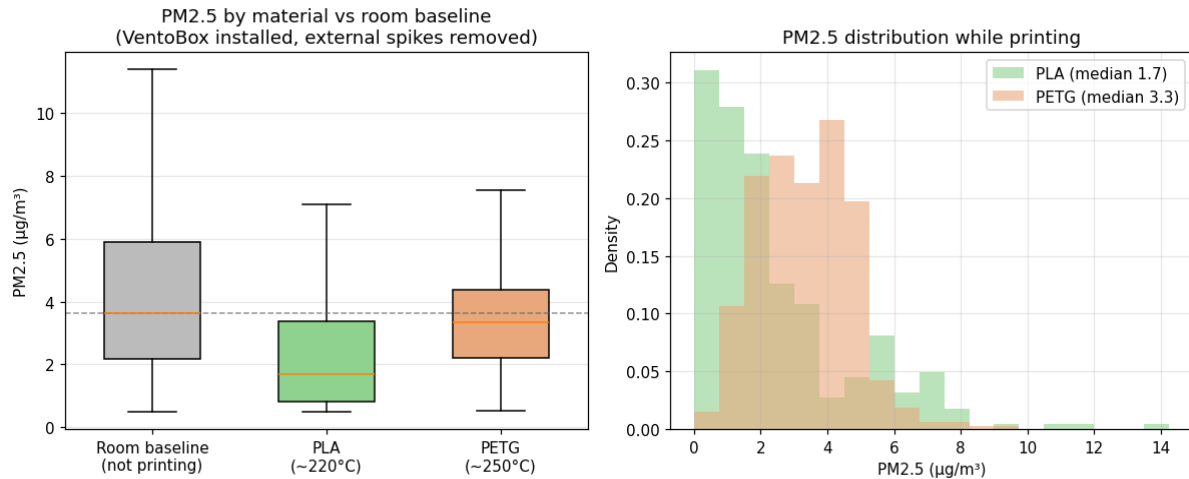


Figure 4. PM2.5 by material versus the room baseline (VentoBox installed, external spikes removed). PETG prints show roughly double the median PM2.5 of PLA, but both stay at or below the non-printing room baseline (dashed line).

The result has two parts. First, PETG does emit more particulates than PLA: its median PM2.5 during printing (3.3 $\mu\text{g}/\text{m}^3$) is about twice that of PLA (1.7 $\mu\text{g}/\text{m}^3$), a highly significant difference ($p \approx 2 \times 10^{-18}$). This matches expectations, since PETG is extruded around 30 °C hotter. Second — and more reassuring — with the VentoBox running, neither material pushed PM2.5 above the normal room baseline (3.6 $\mu\text{g}/\text{m}^3$). PLA printing actually sat about 1.9 $\mu\text{g}/\text{m}^3$ below baseline and PETG about 0.3 $\mu\text{g}/\text{m}^3$ below it.

In other words, the difference between PLA and PETG is real but small in absolute terms, and the filtration is effective enough that printing either material did not measurably degrade the air relative to the room's usual state. The higher PETG figure is best read as “less strongly filtered below baseline,” not as a hazardous level.

6. What Actually Drives the Spikes

Counter to initial expectations, the printer was not the main source of high PM2.5 readings. Of the ten highest 5-minute readings, eight occurred while the nozzle was cold (not printing). The top five events:

Time (UTC)	PM2.5	Printing?
2 Jun 14:30	372 $\mu\text{g}/\text{m}^3$	Yes
2 Jun 14:35	300 $\mu\text{g}/\text{m}^3$	No
2 Jun 14:40	240 $\mu\text{g}/\text{m}^3$	No
24 May 17:55	185 $\mu\text{g}/\text{m}^3$	No
5 Jun 13:10	171 $\mu\text{g}/\text{m}^3$	No

The single highest reading does coincide with printing, but it is immediately followed by even larger readings after the nozzle cooled — suggesting an external event (e.g. cooking, a window opened to a smoggy street) that happened to overlap the end of a print, rather than being caused by it.

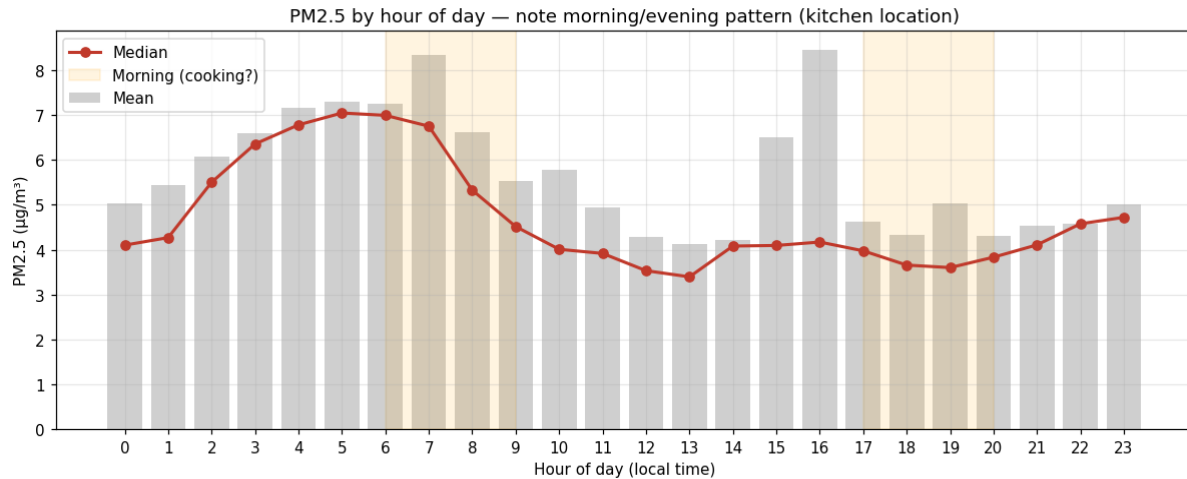


Figure 5. PM2.5 by hour of day (local time). The pronounced early-morning and evening pattern is characteristic of cooking and/or outdoor air ingress — consistent with the sensor's placement near the kitchen and a window — rather than printing, which occurred at varied hours.

Because the sensor sits near the kitchen and a window, two plausible external sources are cooking emissions and outdoor air (including seasonal smog). Distinguishing the two would require an outdoor reference sensor and is left for future work.

7. Gas-Phase Results: What the Particle Sensor Missed

The original study could not analyse VOC and NO_x because a firmware sampling bug prevented the Sensirion gas-index algorithm from calibrating (see Section 8.1). That bug was fixed, and a second fault was later found: the algorithm type constants were transposed, so each channel was processed with the wrong algorithm. After correcting both, a fresh dataset was collected from 25 July to 17 August 2026 (4,287 five-minute samples, of which 321 during active printing). Three days where the VOC channel froze at a constant value were excluded.

The gas data changes the headline conclusion of this report. Where particulates showed no printing signal at all, VOC rises sharply and unambiguously during printing:

Channel	Printing (median)	Idle (median)	Difference	p-value
VOC Index	196	118	+78	7×10^{-14}
PM2.5	3.2 µg/m ³	4.1 µg/m ³	-0.9	5×10^{-5}
NOx Index	1	1	0	—

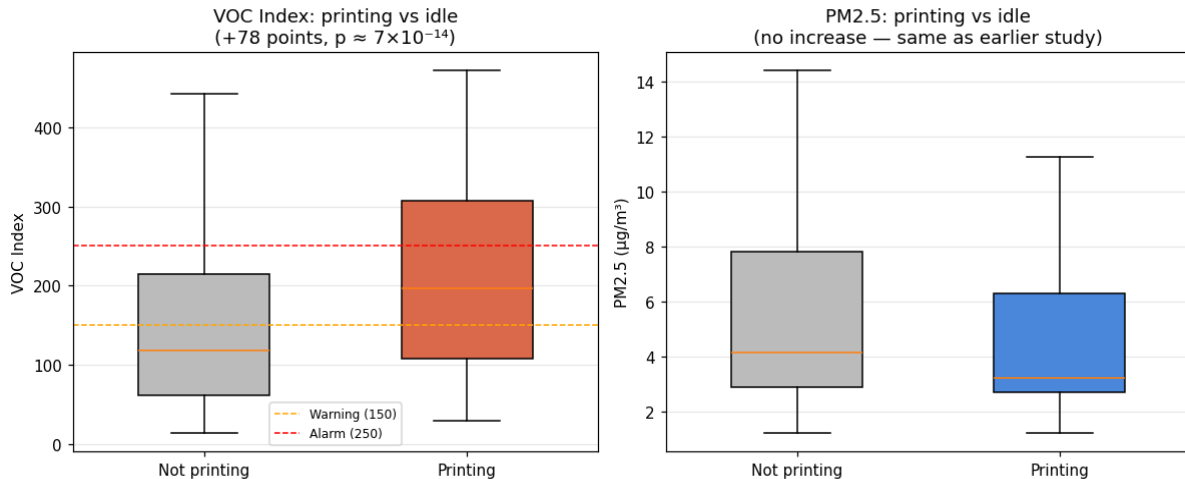


Figure 6. VOC Index and PM2.5 compared between printing and idle periods. The VOC distribution shifts strongly upward during printing, crossing the 150 warning threshold; PM2.5 shows no such shift. Dashed lines mark the VOC warning (150) and alarm (250) levels.

During printing the VOC Index exceeded 150 for 66% of the time and 250 for 36%. Those figures need their counterpart to be read correctly: with the printer switched off the same thresholds were exceeded 39% and 21% of the time. The elevation during printing is real, but the two distributions overlap substantially — this is not a case of clean air punctuated by printing events. The NOx Index remained at its clean-air baseline of 1 throughout, which is expected, as FDM printing does not produce nitrogen oxides.

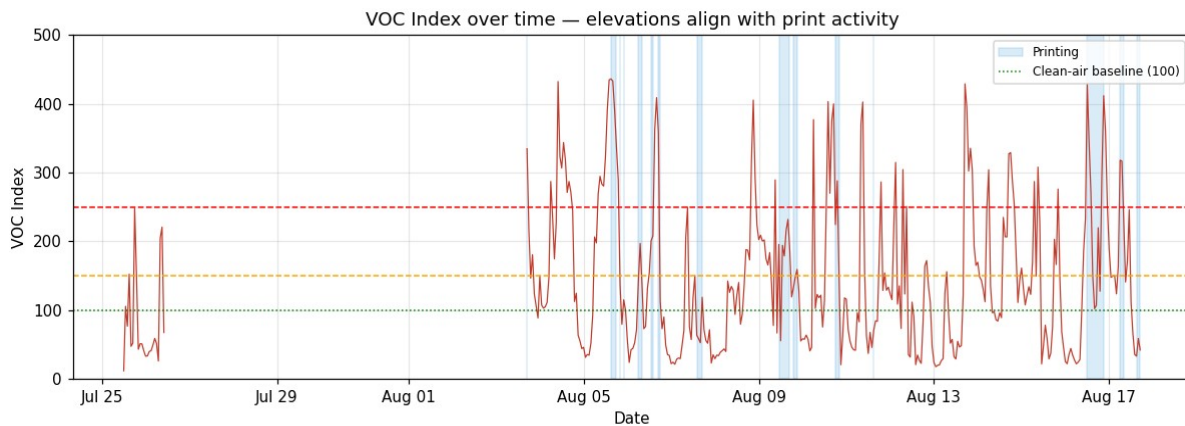


Figure 7. VOC Index over the measurement window, with printing periods shaded. Sustained elevations align with print activity; the green dotted line marks the nominal clean-air baseline of 100.

7.1 How to read these numbers — and how not to

The VOC Index is not a concentration and cannot be compared against health limits. The SGP41 is a metal-oxide sensor: it produces a single signal that responds to oxidisable gases collectively, with no ability to identify which compounds are present or in what quantity. Sensirion's algorithm converts that signal into a 1–500 index in which 100 represents the room's own typical state over roughly the preceding 24 hours. A reading of 200 therefore means "more oxidisable gas than is usual here," not any particular $\mu\text{g}/\text{m}^3$.

This matters because volatile organic compounds are a category of hundreds of substances with vastly different toxicity — benzene and formaldehyde are carcinogens with strict limits;

limonene from citrus is essentially harmless. For that reason the WHO publishes guidelines for individual compounds, not for VOCs as a group, and no threshold in this report corresponds to a health standard. Distinguishing filament off-gassing from cooking, cleaning products, or solvents would require compound-specific instrumentation.

The 150 and 250 thresholds used above were chosen as provisional operational values before any data existed. The data now shows they are too low for this room: the idle distribution alone has a 90th percentile of 373. Thresholds set from the room's own idle distribution (for example warning at 373 and alarm at 432) would flag genuinely unusual conditions rather than ordinary daily variation. This is a monitoring-configuration conclusion, not a health finding.

What can be said from this dataset: printing produces a measurable, statistically robust gas-phase signal that the particulate sensor does not detect at all. What cannot be said: whether that signal represents a health risk. Answering that requires measuring specific compounds against their own guideline values.

7.2 Why this matters for the VentoBox

Section 4 concluded that no PM2.5 benefit from the VentoBox could be demonstrated, while noting that the filter's activated-carbon layer targets gases the sensor could not measure. The gas data confirms that a gas-phase signal from printing genuinely exists and is substantial. It does not, however, quantify the filter's effect: throughout this window the VentoBox ran continuously at 100%, so there is no unfiltered comparison period. Establishing the filter's VOC benefit would require a deliberate experiment — matched prints of the same material with the filter on and off.

7.3 Material comparison remains inconclusive

Filament classification was revised for this phase. The earlier study required PETG to have a bed temperature of 65 °C or above, which misclassified a large group of prints run at 245 °C nozzle with a 60 °C bed. Since nozzle temperature is the reliable discriminator, classification now uses it alone: 195–235 °C for PLA and 235–265 °C for PETG.

In this window PLA accounted for only 39 printing samples against 240 for PETG. Both materials raised VOC substantially above baseline (PLA +100, PETG +76 index points), but the difference between them was not statistically significant ($p = 0.07$) and the sample imbalance is too large to support a conclusion. The finding to take from this phase is that both materials off-gas measurably, not that one is worse than the other.

This also does not overturn the particulate result from Section 5, which was based on a better-balanced sample and found PETG produced roughly twice the PM2.5 of PLA. Gases and particles behave differently, and there is no reason to expect the same ranking.

7.4 Device reliability after the fixes

Stability improved markedly. A cold solder joint on the gas sensor's power pin — which had been intermittently cutting power and resetting the algorithm baseline — was repaired. Combined with the firmware changes, restarts fell from 29 in 25 days to just two in the month following the fix, both RTC watchdog resets.

8. Limitations & Future Work

8.1 Gas-phase data: the original fault

During the first phase, VOC and NO_x values remained pinned at their initialisation values (VOC = 1, NO_x = 100) with zero variance, even across a continuous 30-hour window. Three separate faults were eventually found. First, the Sensirion gas-index algorithm requires its processing function to be called at a steady 1 Hz, but the original firmware called it irregularly, so the algorithm never built its internal baseline. Second, a cold solder joint on the sensor's power pin intermittently cut power, resetting whatever baseline had accumulated. Third, the algorithm type constants were transposed in the constructor, so each channel was processed with the other's algorithm — which is why the two baselines appeared swapped once data finally began to vary.

All three were corrected: the gas sensor is now sampled on a dedicated, unconditional 1-second timer; the power pin was resoldered; and the algorithm types were fixed. The resulting data is analysed in Section 7. A lesson for future work: the raw sensor signals should also be logged alongside the computed indices, since the index algorithm is stateful and non-invertible — with only the processed values stored, the mislabelled data from the first phase could not be recovered.

8.2 Device reliability

Diagnostic logging (free heap, RSSI, CPU temperature, boot reason) was used to characterise stability. Over the period the device restarted on 29 occasions. Boot-reason analysis showed an early cluster (18–19 May) of power-cycles and WiFi-loss restarts during active development, followed by periodic interrupt-watchdog resets every 1–3 days. Free heap was rock-stable throughout (never below 219 KB, drift ≈ -5 bytes/hour), ruling out a memory leak; the watchdog resets are attributed to occasional blocking on the I2C bus or network calls. The revised firmware adds an I2C timeout, bounded reconnect attempts, and regular watchdog servicing to address this.

8.3 Other limitations

- No outdoor reference sensor, so indoor-vs-outdoor attribution of spikes is inferred, not measured.
- Filament type (PLA vs PETG) was not logged per-print, so material-specific emission differences could not be isolated.
- 5-minute averaging smooths short transient events; high-resolution capture would better resolve brief emission peaks.

9. Conclusions

- Indoor PM_{2.5} in the flat was good overall, below the WHO 24-hour guideline 97% of the time.
- Printing PLA or PETG barely raised PM_{2.5} above the room baseline — even without filtration (+0.6 $\mu\text{g}/\text{m}^3$, not significant). A clear PM_{2.5} benefit from the VentoBox could not be isolated, because the printer added so little to begin with.
- The VentoBox should not be dismissed. The gas-phase data confirms a substantial VOC signal from printing — precisely what an activated-carbon filter targets — but the filter ran continuously throughout that phase, so its benefit remains unquantified. A matched on/off experiment is the obvious next step.

- PETG produced about twice the median PM_{2.5} of PLA while printing (3.3 vs 1.7 µg/m³), consistent with its higher extrusion temperature — but with the VentoBox running, both materials stayed at or below the room baseline.
- The largest pollution events originated outside the printer, consistent with cooking and outdoor air given the sensor location.
- Gas-phase monitoring, once working, revealed what the particulate sensor could not: printing raises the VOC Index by roughly 78 points above the idle median. Both PLA and PETG off-gas measurably. Because the index is relative rather than a concentration, this establishes that printing alters the air — not that it is hazardous; the provisional 150/250 thresholds proved too low for this room and should be reset from its own idle distribution.

Data collected 16 May – 11 June 2026. Analysis based on 6,781 five-minute samples of sensor data time-aligned with 4,652 printer status records. Hardware: ESP32 + Sensirion SPS30/SGP41; pipeline: MQTT → Telegraf → InfluxDB → Grafana.